

ME 517: Advanced Analytical Techniques for Engineers

Course Code	ME 517
Course Name	Advanced Analytical Techniques for Engineers
L-T-P-C	3-1-0-4
Prerequisites	None
Intended for	UG/MTech/MS/PhD in engineering streams core for M.Tech students in Fluid and Thermal Engineering
Distribution	Elective
Approval	30th Senate (Feb 2021), in the M.Tech Fluid and Thermal Engineering (FTE) curriculum; 40th BoA (May 2021); 35th Senate (Jun 2022), in the M.Tech Fluid and Thermal Engineering curriculum

Course Contents

- **Integral Transforms**
Fourier transform for solution of ODEs and PDEs, Laplace transform for solution of ODEs and PDEs. [6 Lectures]
- **Partial Differential Equations**
solution of linear PDEs including special cases of heat conduction equation and Navier–Stokes equation, solution of PDEs in cylindrical and spherical coordinate systems. [10 Lectures]
- **First-order integral equations**
Fredholm, Volterra and Wiener-Hopf equation, power series solution for integral equations, integral equations as a generalization of eigenvalue equations and connection to inverse problems. [6 Lectures]
- **Tensors**
Einstein notation, tensor transformations, tensor fields and tensor calculus, integral theorems. [6 Lectures]
- **Complex analysis**
functions of a complex variable, limits and continuity, analytic functions, complex exponents, contour integrals, Cauchy integral formula, complex series, Cauchy's residue theorem, singularity, zeros and poles, improper integrals in Fourier analysis, application to plane fluid flow and signal processing. [8 Lectures]
- **Linear Algebra**
spaces and subspaces, positive definiteness and singular value decomposition (best basis), transformations and change of basis, pseudoinverse, eigenvalues and eigenvectors, introduction to multigrid, Krylov subspaces and conjugate gradients methods for solving large linear systems. [10 Lectures]
- **Statistical Methods**
Probability, random variables, discrete and continuous probability distributions, autocorrelation and cross correlation, Monte-Carlo method for the solution of diffusion equation, radiative transfer equation and collimated beam radiation problem; linear regression and curve fitting. [10 Lectures]

Text Books

1. B. Dasgupta, **Applied Mathematical Methods**, Pearson Education, 2006.
2. Sheldon M Ross, **Probability and Statistics for Engineers and Scientists**, 5th Edition, Academic Press, 2014.

References

1. M D. Greenberg, **Advanced Engineering Mathematics**, 2nd Edition, Pearson India, 2007.
2. E. Kreyszig, **Advanced Engineering Mathematics**, International 10th Revised Edition, John Wiley and Sons, 2015.
3. G. Strang, **Linear Algebra and Its Applications**, Thomson, Brooks/Cole, 2006
4. P. V. O'Neil, **Advanced Engineering Mathematics**, CENGAGE Learning, 2011.
5. R. L. Thomas and G. B. Finney, **Calculus and Analytic Geometry**, 11/e Addition-Wisley Reading, 2010.
6. M. L. Boas, **Mathematical Methods in the Physical Sciences**, 3rd Ed, Wiley India, 2009.
7. A. D. Polyanin and A. V. Manzhirov, **Handbook of Integral Equations**, 2nd Edition, CRC Press, 2008.